

Lab 3: OSI and TCP/IP model and implementing a LAN network

Objective

- Name the seven layers of the OSI model, in order. Use a mnemonic.
- Describe the characteristics, functions and keywords relating to each layer.
- Describe the packaging units used to encapsulate each layer.
- Name the physical devices or components that operate at each layer.
- Describe the four layers of the TCP/IP model.
- Relate the seven layers of the OSI model to the four layers of the TCP/IP model.
- Name the primary TCP/IP protocols and utilities that operate at each layer.
- Implementation of a basic LAN network between two computers using Ethernet cable

Background

This lab will help to develop a better understanding of the seven layers of the OSI model. Specifically as they relate to the most popular functioning networking model in existence, the TCP/IP model. The Internet is based on TCP/IP. TCP/IP has become the standard language of networking. However, the seven layers of the OSI model are the ones most commonly used to describe and compare networking software and hardware from various vendors. It is very important to know both models and be able to relate or map the layers of one to the other. An understanding of the TCP/IP model and the protocols and utilities that operate at each layer is essential when troubleshooting.

Steps

1. List the seven layers of the OSI model from the top to the bottom. Give a mnemonic word for each layer that can help you remember it. Then list the keywords and phrases that describe the characteristics and function of each.

Layer #	Name	Mnemonic	Key Words and Description of Function

2. List the seven layers of the OSI model and the encapsulation unit used to describe the data grouping at each layer. Also list the networking devices that operate at each layer, if applicable.

Layer #	Name	Encapsulation Unit or Logical Grouping	Devices or Components that Operate at this Layer

3. Use the table below to compare the OSI layers with the TCP/IP protocol stack. In column two, indicate the proper name for each of the seven layers of the OSI model corresponding to the layer number. List the TCP/IP layer number and its correct name in the next two columns. Also list the term used for the encapsulation units, the related TCP/IP protocols and utilities that operate at each TCP/IP layer. More than one OSI layer will be related to certain TCP/IP layers.

OSI comparison with TCP/IP Protocol Stack

OSI #	OSI Layer Name	TCP/IP #	TCP/IP Layer Name	Encapsulation Units	TCP/IP Protocols at Each TCP/IP Layer	TCP Utilities
7						
6						
5						
4						
3						
2						
1						

Implementing a LAN network

In this part of lab a basic LAN network is implemented between two computers using Ethernet cable and data is transferred.

Component required:

- Two computers
- Ethernet cable

Steps:

Following steps are performed

- i. Connect Ethernet cable between two computers
- ii. Turn on firewall
- iii. Assign IP addresses of both computers
- iv. Create a Home group
- v. Joining home group
- vi. Sharing drives/folders

Lab Tasks

Post Lab Task 1:

Implement a basic LAN network between two computers using Ethernet cable and explain all steps in detail. Draw a flow chart also to explain all steps.